Project Development Phase

Model Performance Test – Cyber Security With AI

Project Name	AI-Enhanced Intrusion Detection System

Model Performance Testing (VAPT):

Project team has filled the following information for VAPT testing for the target system.

S.No.	Parameter	Values
1.	**Information Gathering**	Footprinting – Target: Web application traffic classification (simulated network environment) Reconnaissance – CICIDS2017 dataset analyzed; 78 network flow features identified; attack patterns mapped for Brute Force, SQL Injection, and XSS
2.	**Scanning the Target**	Scanning Info – Network flow parameters scanned: Flow Duration, Total Fwd Packets, Total Bwd Packets, Total Length of Fwd Packets Risk Factors – High-risk: SQL Injection (payload length > 400 bytes), Brute Force (high packet count, long duration), XSS (moderate forward packets)
3.	**Gaining Access**	Access Process – REST API endpoint /predict receives JSON payloads; model classifies traffic in real-time Vulnerability Found – Web Attack vectors detected: Brute Force, SQL Injection, XSS; classified with ~98.9% accuracy
4.	**Maintaining Access – Automation (AI Implementation)**	AI Tools Used – Random Forest Classifier (Scikit-Learn), SMOTE (imbalanced-learn), Flask REST API Automation Implemented – Continuous packet flow monitoring via Live Threat Log; automated real-time classification pipeline; WAF integration via JSON API
5.	**Covering Tracks & Report**	Vulnerability Risk Factors – SQL Injection (Critical), Brute Force (High), XSS (Medium), BENIGN traffic (No risk) VAPT Report – Detection system achieves 98.9% accuracy; model optimized from 78 to 4 features; all 3 attack classes successfully detected in testing